A CRM APPLICATION FOR THE CHAI SHOP & CO

3. PROJECT DESIGN PHASE

3.1 Introduction

The design phase converts requirements into a technical structure.

The proposed CRM can follow a three-layer architecture:

Presentation
↓
Application/Business
↓
Database Layer

3.2 Presentation Layer

This is the user interface.

Main screens:

- Login
- Dashboard
- Customer Registration
- Customer Profile
- Customer Search
- Orders
- Loyalty
- Feedback
- Complaints
- Offers
- Reports
- User Management

3.3 Application Layer

The application layer handles business rules.

Examples:

- Validate customer data.
- Calculate loyalty points.
- Prevent unauthorized operations.
- Create customer records.
- Update customer profiles.
- Record feedback.
- Generate reports.
- Manage campaigns.

3.4 Database Layer

The database stores application information.

Potential tables include:

- Users
- Customers
- Products
- Orders
- Order Items
- Loyalty Accounts
- Loyalty Transactions
- Feedback
- Complaints
- Campaigns
- Offers
- Customer Offers
- Interactions

3.5 Customer Table

Field	Description
Customer_ID	Unique customer identifier
Name	Customer name
Mobile	Contact number
Email	Email address
Preference	Product preference
Created Date	Registration date
Status	Active/Inactive

3.6 Order Table

Field	Description
Order ID	Unique order number
Customer_ID	Linked customer
Order Date	Date/time
Total	Total order value
Order Type	Dine-in/Takeaway/Online
Status	Order status

3.7 Feedback Table

Field	Description
Feedback ID	Unique ID
Customer_ID	Customer

Field	Description
Order ID	Related order
Rating	Rating
Comment	Feedback
Date	Submission date
Status	Open/Closed

3.8 Loyalty Table

Field	Description
Loyalty ID	Unique ID
Customer_ID	Customer
Points	Current points
Total Earned	Total earned
Total Redeemed	Total redeemed

3.9 Login Screen

The login page contains:

- Username/email
- Password
- Login button
- Forgot password option

3.10 Dashboard

The dashboard displays summarized information:

- Total customers
- Today's orders
- New customers
- Pending complaints
- Loyalty activity
- Recent feedback
- Active campaigns

3.11 Customer Profile Screen

The customer profile should provide a 360-degree view of relevant customer information.

Tabs can include:

Profile | Orders | Loyalty | Feedback | Complaints | Interactions

This centralized approach reflects a common CRM design principle of bringing customer information and interactions together.

3.12 Navigation

A left-side or top navigation menu can contain:

1. Dashboard
2. Customers
3. Orders
4. Loyalty
5. Feedback
6. Campaigns
7. Reports
8. Settings
9. Logout

3.13 Customer Registration Flow

Start

↓

Employee enters customer details

↓

System validates data

↓

Check for duplicate customer

↓

If valid → Save customer

↓

Generate Customer ID

↓

Display confirmation

↓

End

3.14 Loyalty Flow

Order Completed

↓

Identify customer

↓

Calculate eligible points

↓

Add points to account

↓

Update loyalty balance

↓

Display updated balance

3.15 Complaint Flow

Complaint Submitted

↓

Create complaint

↓

Assign employee

↓

Investigate issue

↓

Update resolution

↓

Notify/record outcome

↓

Close complaint

3.16 Design Goals

The system design emphasizes:

- Simplicity
- Maintainability
- Security
- Scalability
- Accurate data
- Fast access